

VANSH RANA

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EDUCATION

Chandigarh University — Bachelor of Engineering, Computer Science

Mohali, Punjab | Expected May 2028

SGPA: 8.1/10 · Coursework: Data Structures & Algorithms, OOP, DBMS, OS, System Design

Army Public School, Patiala — Class XII (CBSE): 87% | Class X (CBSE): 89%

2022 – 2024

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL **Frameworks:** React.js, Node.js, Express.js, FastAPI, Scikit-learn, WebRTC, Solidity

Databases: MongoDB, PostgreSQL **Tools & Concepts:** Git, REST APIs, JWT, DSA, OOP, System Design, OS, Blockchain, MILP Optimization

EXPERIENCE & ACHIEVEMENTS

Full Stack Engineer Intern — ThinkNEXT Technologies Pvt. Ltd.

Mohali, India | May 2026 – Present

- Accelerated delivery of 3+ production features by ~20% as measured by sprint velocity by architecting RESTful APIs and reusable React component libraries under the MERN stack (MongoDB, Express.js, React, Node.js).
- Improved codebase maintainability by applying MVC architecture and input validation middleware to enterprise-scale web modules, reducing bug recurrence across a 6-week on-site engagement.

1st Place — **BuildX National Hackathon** · NSUT Delhi, Sponsored by TRAE AI

Feb 2026

- Won ₹50,000 cash prize and secured 1st among 100+ national teams as measured by algorithmic accuracy, financial viability, and innovation by leading Team APEX ZERO to deliver a live ML-powered IPL Roster Optimizer within 24 hours.

OPEN SOURCE CONTRIBUTIONS

[fastify-param-schema-validation](#) · Node.js · Fastify · JSON Schema · fastify-plugin

Open Source | 2026

- Designed, published, and open-sourced a plugin officially indexed in the Fastify framework ecosystem — leverages the onRoute lifecycle hook to audit every route's URL parameters against its JSON schema at server startup.
- Broke through Fastify's native encapsulation using fastify-plugin to engineer a global, zero-runtime-overhead middleware that enforces schema completeness across the entire server instance automatically.
- Enforced a fail-fast boot sequence by throwing a custom FST_ERR_SCH_VALIDATION_BUILD exception on parameter-schema mismatch, guaranteeing misconfigured APIs can never slip into a live production environment.

PROJECTS

AI-Driven IPL Roster Optimizer (Apex Zero) · React · FastAPI · Scikit-learn · MILP/PuLP · Python · Recharts

Feb 2026

- Achieved 85%+ player performance prediction accuracy on a holdout IPL dataset by engineering a Scikit-learn ML pipeline (XGBoost/Random Forest) trained on 200+ career metrics across 400+ IPL players.
- Cut manual squad-building time by ~70% by implementing MILP (PuLP) for optimal 25-player squad selection under ₹80Cr salary caps with Recharts dashboards surfacing undervalued, overpaid, and high-potential players.
- Deployed across Render (FastAPI backend) and Netlify (Vite/React frontend) with environment variable baking and unified repo structure, enabling zero-downtime production releases.

DataPassport · Next.js · Express/TypeScript · Solidity · Blockchain · MongoDB · Gemini AI

Mar 2026

- Eliminated tampering risk for 100+ AI training assets by anchoring SHA-256 fingerprints to Base Sepolia blockchain via Solidity smart contracts, verified through on-chain hash validation.
- Reduced manual audit overhead by ~60% by integrating Gemini AI risk-scoring and MetaMask wallet auth to auto-classify dataset risk tiers and verify provenance across the full dataset lifecycle.
- Engineered a URL-scanning pipeline to evaluate external dataset sources for credibility, surfacing actionable trust scores to researchers before committing datasets to on-chain notarization.

CampusCoin · MERN Stack · PWA · JWT · Socket.io · bcrypt · UPI Integration

Jan 2026

- Enabled peer-to-peer commerce for 500+ students by architecting a MERN PWA campus marketplace with listings, skill-swap, and profile modules secured with JWT + bcrypt authentication.
- Engineered real-time task notifications via Socket.io with sub-200ms delivery latency and a virtual CampusCoin economy with anti-abuse mechanics governing all peer-to-peer transactions.
- Integrated UPI payment support (PhonePe, GPay, BHIM, Paytm) with a fraud-prevention layer, enabling frictionless real-money settlement alongside the in-platform coin economy.

EXTRACURRICULARS & INTERESTS

Former U-16 State-Level Cricket Player (Punjab) · Fitness & Strength Training · Self-directed learner — applies structured, first-principles approach to mastering new technologies · Competitive hackathon participant (national level, 2x podium finish)